CS 4530
Fundamentals of Software Engineering

Module 15: Ethics in Software Engineering

Adeel Bhutta, Rob Simmons and Mitch Wand
Khoury College of Computer Sciences

Goal: to make software an engineering discipline

Software engineering concerns the

- design
- construction,
- and maintenance
- ➢ of large programs
- ➢ over time.

Programs are not morally neutral things

- programs can have good and ill effects for people
- programs represent and encode human values

Programs can have good and ill effects

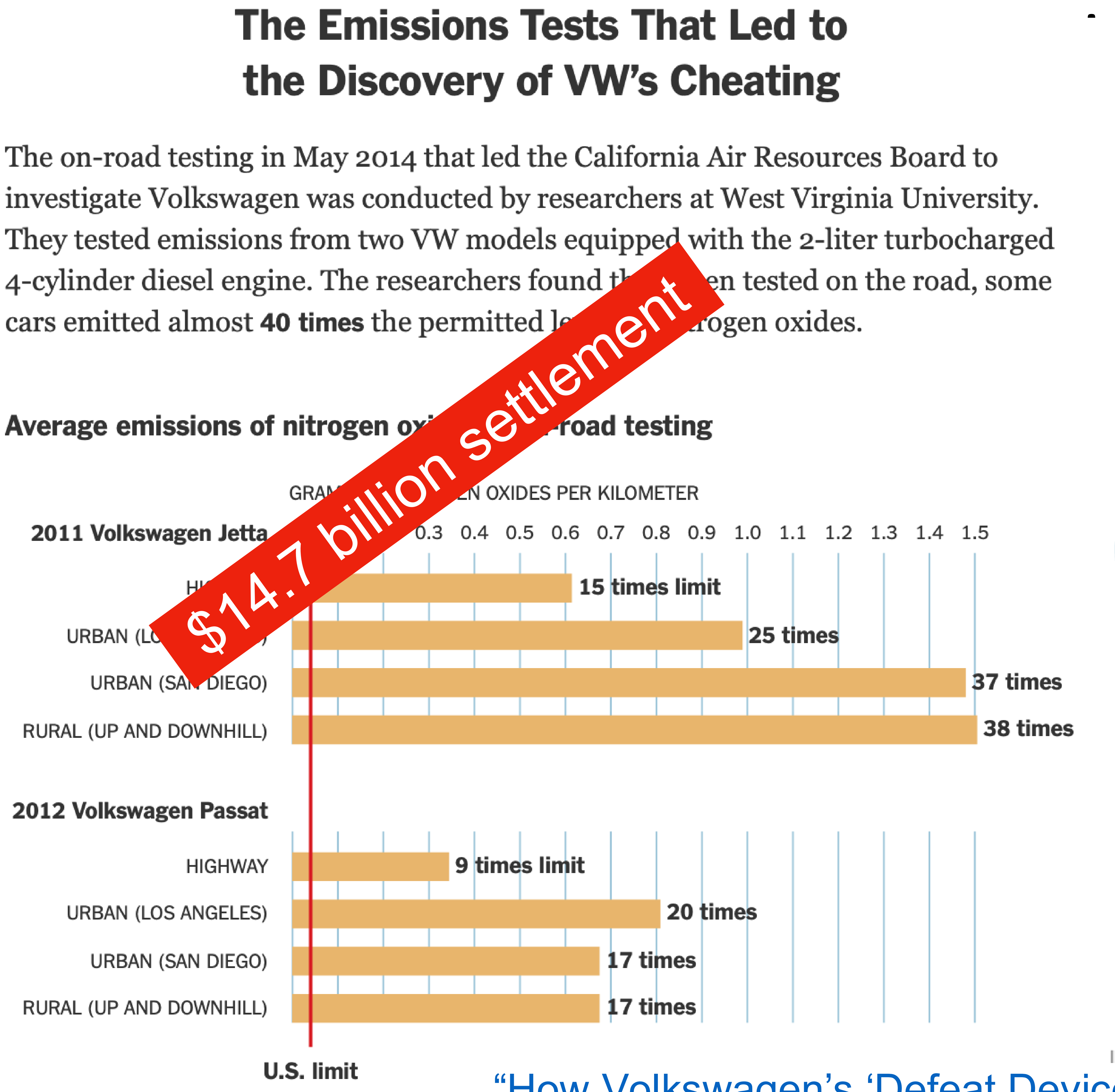
Badly-engineered software can kill people

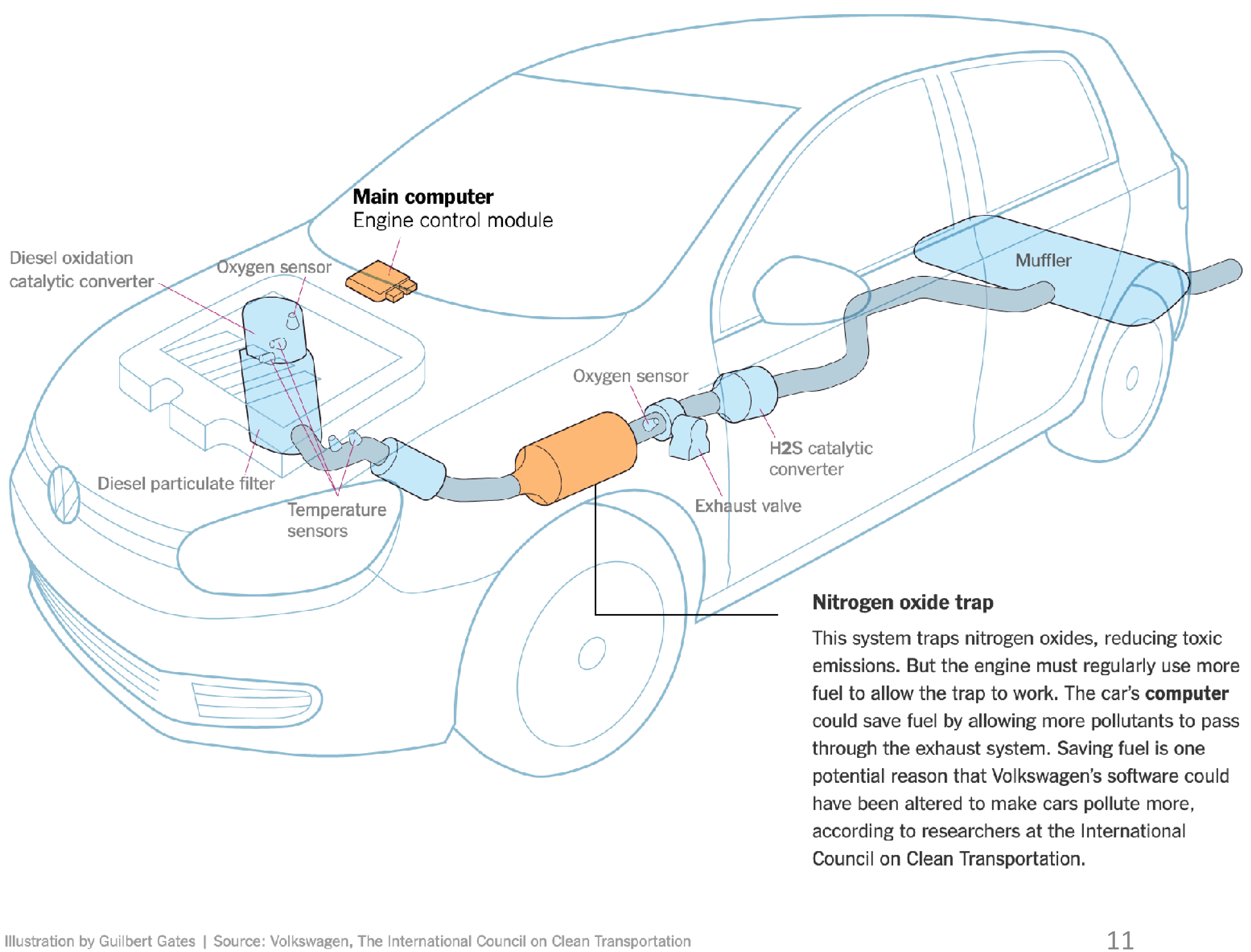
- Therac-25 (1985-1987)
- Bug in software caused 100x greater exposure to radiation than intended
- At least 6 died
- Likely far more suffered: deaths occurred over a period of 2 years!
- Weak accountability in manufacturer’s organization

“Therac-25” by Catalina Márquez, Wikimedia commons, CC BY-SA 4.0

https://ethicsunwrapped.utexas.edu/case-study/therac-25

Criming Corporate Bosses Need Software Engineers





[“How Volkswagen’s ‘Defeat Devices’ Worked” By Guilbert Gates, Jack Ewing, Karl Russell and Derek Watkins]

Criming Corporate Bosses Need Software Engineers

Volkswagen executives get prison time in 'Dieselgate' scandal

Volkswagen has admitted that some of its engineers installed emissions test cheating software in diesel-powered vehicles

By **Kevin Williams** Updated May 27, 2025

Is This A Good Idea? It Can Depend on What Question You're Asking

There are deaths here attributed to software bugs!

List of Tesla Autopilot crashes

Article Talk

From Wikipedia, the free encyclopedia

Tesla Autopilot, a Level 2 advanced driver assistance system (ADAS), was released in October 2015 and the first fatal crashes involving the system occurred less than one year later. The fatal crashes attracted attention from news publications and United States government agencies, including the National Transportation Safety Board (NTSB) and National Highway Traffic Safety Administration (NHTSA), which has argued the Tesla Autopilot death rate is higher than the reported

Tesla will tell you that more lives were saved than if the software hadn't existed

Is This A Good Idea? It Can Look Different To Different People

One mark of an exceptional engineer is the ability to understand how products can advantage and disadvantage different groups of human beings

Engineers are expected to have technical aptitude, but they should also have the discernment to know when to build something and when not to

Demma Rodriguez
Head of Equity Engineering
Google

Quote: “Software Engineering at Google: Lessons Learned from Programming Over Time,” Wright et al, 2020 (O’Reilly)

Is This A Good Idea? It Can Look Different To Different People

"Earlier this week, X announced it will soon roll out a new function, allowing blocked accounts to still view public posts by users who have blocked them… One commenter wrote, "Big day for stalkers and harassers," while another stated, "That's not blocking. It's supporting stalking.""

https://tech.yahoo.com/apps/articles/bluesky-app-trends-elon-musk-172408742.html

Programs represent and encode human values

ML (&LLMs&etc) Reinforce, "Launder" Bias

- The COMPAS sentencing tool discriminates against black defendants

	ALL	**WHITE** DEFENDANTS	**BLACK** DEFENDANTS
Labeled **High Risk**, But **Didn't** Re-Offend	**32%**	**23%**	**44%**
Labeled **Low Risk,** Yet **Did** Re-Offend	**37%**	**47%**	**28%**

Analysis of Broward County data: [“How We Analyzed the COMPAS Recidivism Algorithm” by Larson et al.]

ML (&LLMs&etc) Reinforce, “Launder” Bias

According to Wilson & Caliskan:

- human biases lead to people with names identified as “white” getting hired more
- system trained on who got hired in the past

Machine learning algorithms can be great at teasing out how we’re actually making decisions (sometimes badly!)

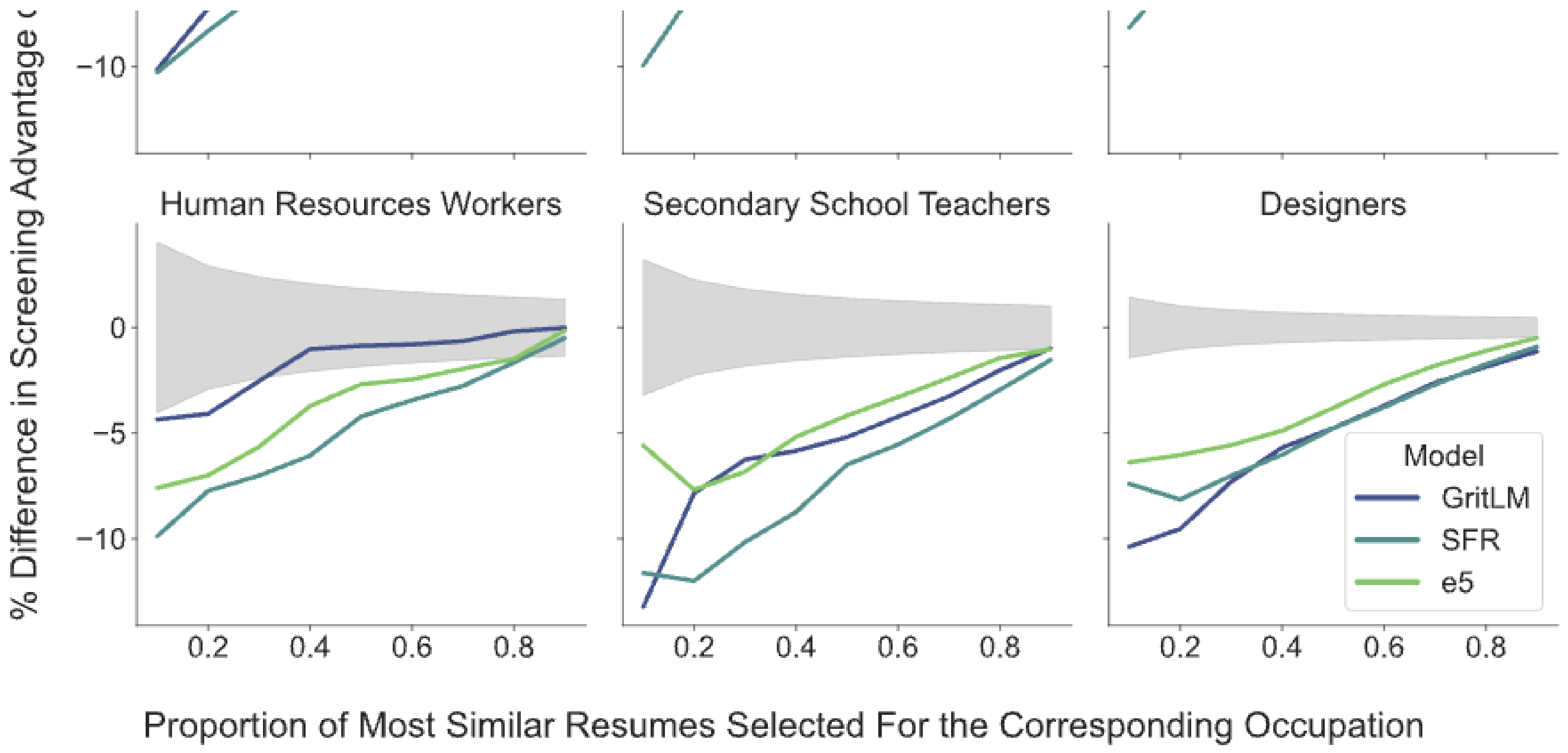


Figure 9: Resumes with White male names are preferred in 100% of tests; those with Black male names are preferred in 0%. Gray regions indicate disparities which are not significantly different from zero (0% of tests).

Kyra Wilson & Aylin Caliskan, “Gender, Race, and Intersectional Bias in Resume Screening via Language Model Retrieval”
https://ojs.aaai.org/index.php/AIES/article/view/31748

Code is Not Morally Neutral

“Here we illustrate the point that algorithms themselves can be the source of bias with the example of collaborative filtering algorithms for recommendation and search.”

Bias in collaborative filtering, Catherine Stinson, AI and Ethics Volume 2, 2022

https://link.springer.com/article/10.1007/s43681-022-00136-w

Stable Matching and the NRMP

"When residencies were first introduced, around 1900, hospitals began competing with one another to secure the best residents as early as possible. By the 1940s, positions were being offered in the third-year of medical school. Students were making career decisions without adequate exposure to their options, and hospitals were making hiring decisions with little data."

https://pmc.ncbi.nlm.nih.gov/articles/PMC3399603/

Stable Matching and the NRMP

- 1952 National Resident Matching Program — big centralized assignment of residents to programs.
- *Stable Matching* is a property of weighted graphs: if Jimmy gets assigned to Mass General but would prefer Emory University, Emory prefers *everyone* assigned to them over Jimmy. (No defections.)
- Algorithm naturally has “proposers” and “acceptors,” and generalizing how applications generally work, natural to make potential applicants the proposers.
- Multiple stable matchings usually exist. That “natural” choice leads to the *best* outcomes for hospitals, and the *worst* outcomes for students.

Stable Matching and the NRMP

American Economic Review

ISSN 0002-8282 (Print) | ISSN 1944-7981 (Online)

About the *AER* ⌄ Articles and Issues ⌄ Information for Authors and Reviewers ⌄

The Redesign of the Matching Market for American Physicians: Some Engineering Aspects of Economic Design

Alvin E. Roth

Elliott Peranson

AMERICAN ECONOMIC REVIEW
VOL. 89, NO. 4, SEPTEMBER 1999
(pp. 748–780)

https://www.aeaweb.org/articles?id=10.1257/aer.89.4.748

Fixes several problems

- Switches proposer/acceptor roles
- *Handles the reality of married couples that want to match in the same program or city*

Even Database Schema are Not Morally Neutral!

Lillie Sep 25, 2017 @ 3:36pm

Transgender dysphoria and steam username

NOTE: This message is mostly for the moderators since there's not a contact us option. I won't respond to hate messages.

So I've had this account for years and built up so much on it. Problem is, everytime i log in, I see my dead name. Yes, I can change my name that people see me by but the

https://steamcommunity.com/discussions/forum/1/1496741765144289150/

Falsehoods Programmers Believe About Names

1. People have exactly one canonical full name.
2. People have exactly one full name which they go by.
3. People have, at this point in time, exactly one canonical full name.
4. People have, at this point in time, one full name which they go by.
5. People have exactly N names, for any value of N.
6. People's names fit within a certain defined amount of space.
7. People's names do not change.
8. People's names change, but only at a certain enumerated set of events.
9. People's names are written in ASCII.
10. People's names are written in any single character set.
11. People's names are all mapped in Unicode code points.
12. People's names are case sensitive.
13. People's names are case insensitive.
14. People's names sometimes have prefixes or suffixes, but you can safely ignore those.
15. People's names do not contain numbers.
16. People's names are not written in ALL CAPS.
17. People's names are not written in all lower case letters.

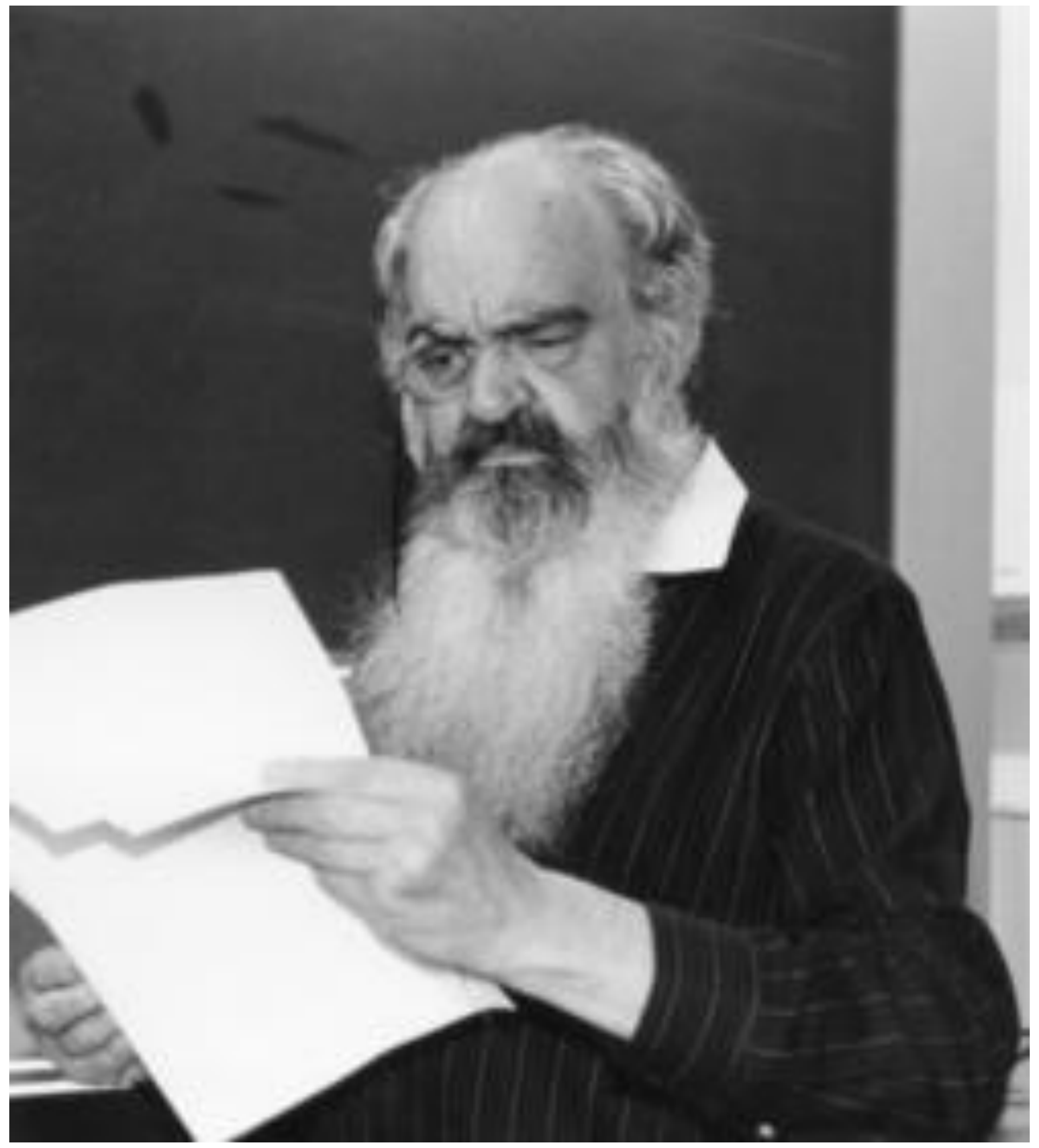
https://www.kalzumeus.com/2010/06/17/falsehoods-programmers-believe-about-names/

The Purpose of Code is What it Does

Systems: The Purpose of a System is What It Does

When trying to understand systems, one really eye-opening and fundamental insight is to realize that *the machine is never broken*. What I mean by this is, when observing the outcomes of a particular system or institution, it's very useful to start from the assumption that the outputs or impacts of that system are precisely what it was designed to do — whether we find those results to be good, bad or mixed.

https://www.anildash.com/2024/05/29/systems-the-purpose-of-a-system/



Stafford Beer, "management cybernetics"

https://en.wikipedia.org/wiki/Stafford_Beer

The Purpose of Code is What it Does

Australian Government
Australian Institute of Family Studies

Discovering
what works
for families

Harms associated with loot boxes, simulated gambling and other in-game purchases in video games: a review of the evidence

Nancy Greer, Cailem Murray Boyle and Rebecca Jenkinson

Australian Gambling Research Centre
Australian Institute of Family Studies

June 2022

https://www.classification.gov.au/sites/default/files/documents/agrc_literature_review_final_20220906_accessible.pdf

There are SE-level mitigations for some of these risks

- Form a diverse team
  - “Tell me you don’t have any ____ on your team without telling me…”
  - People from diverse backgrounds bring different experiences and different perspectives
- Consider human values throughout the project
- Be intentional (& flexible) about stakeholders
- Rely on standards when possible
- Monitor actual usage & misusage, user feedback
  - This is basically the philosophy of continuous delivery, but remember the McNamara fallacy (what gets measured is what gets done)

What values might our software promote or diminish?

- Human rights - Inalienable, fundamental rights to which all people are entitled
- Accessibility - Making all people successful users of the technology
- Justice - Procedural justice (process is fair) + distributive justice (outcomes are fair)
- Privacy - An individual's agency in determining what information about them is shared
- Human welfare - Physical, material and psychological well-being

Value Sensitive Design @ Khoury

"Value constraints," can be framed as negative user stories:

- "As a parent of a patient, I want my child's medical records *not* to be disclosed to unauthorized persons.

Identifying and Filtering Stakeholders

Direct Stakeholders

The sponsor (your employer, etc.)

Members of the design team

Demographically diverse users

- Races and ethnicities, men and women, LGBTQIA, differently abled, US vs. non-US, …

Special populations

- Children, the elderly, victims of intimate partner violence, families living in poverty, the incarcerated, indigenous peoples, the homeless, religious minorities, non-technology users, celebrities

Roles

- Content creators, content consumers, power users, …

Indirect Stakeholders

Bystanders

- Those who are around your users
- E.g. pedestrians near an autonomous car

“Human data points”

- Those who are passively surveilled by your system

Civil society

- E.g. people who aren’t on social media are still impacted by disinformation
- People who care deeply about the issues or problem being addressed

Those without access

- Barriers include: cost, education, availability of necessary hardware and/or infrastructure, institutional censorship…

https://ethics.acm.org/code-of-ethics/software-engineering-code/

ACM Software Engineering Code of Ethics

. PUBLIC – Software engineers shall act consistently with the public interest.

2. CLIENT AND EMPLOYER – Software engineers shall act in a manner that is in the best interests of their client and employer consistent with the public interest.

3. PRODUCT – Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.

4. JUDGMENT – Software engineers shall maintain integrity and independence in their professional judgment.

5. MANAGEMENT – Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.

6. PROFESSION – Software engineers shall advance the integrity and reputation of the profession consistent with the public interest.

7. COLLEAGUES – Software engineers shall be fair to and supportive of their colleagues.

8. SELF – Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.

Does this code change developer behavior?

Does ACM's Code of Ethics Change Ethical Decision Making in Software Development?

Andrew McNamara
North Carolina State University
Raleigh, North Carolina, USA
ajmcnama@ncsu.edu

Justin Smith
North Carolina State University
Raleigh, North Carolina, USA
jssmit11@ncsu.edu

Emerson Murphy-Hill
North Carolina State University
Raleigh, North Carolina, USA
emerson@csc.ncsu.edu

ABSTRACT

Ethical decisions in software development can substantially impact end-users, organizations, and our environment, as is evidenced by recent ethics scandals in the news. Organizations, like the ACM, publish codes of ethics to guide software-related ethical decisions. In fact, the ACM has recently demonstrated renewed interest in its code of ethics and made updates for the first time since 1992. To better understand how the ACM code of ethics changes software-

The first example is the Uber versus Waymo dispute [26], in which a software engineer at Waymo took self-driving car code to his home. Shortly thereafter, the engineer left Waymo to work for a competing company with a self-driving car business, Uber. When Waymo realized that their own code had been taken by their former employee, Waymo sued Uber. Even though the code was not apparently used for Uber's competitive advantage, the two companies settled the lawsuit for $245 million dollars.

TLDR: No

Where does this leave us?

- **So that we can sleep at night**
  - Consider the different ways that our software may impact others
  - Consider the ways in which our software interacts with the political, social, and economic systems in which we and our users live
  - Follow best practices, and actively push to improve them
  - Encourage diversity in our development teams
  - Engage in honest conversations with our co-workers and supervisors to explore possible ethical issues and their implications.

Also… you just don't always get to sleep at night.